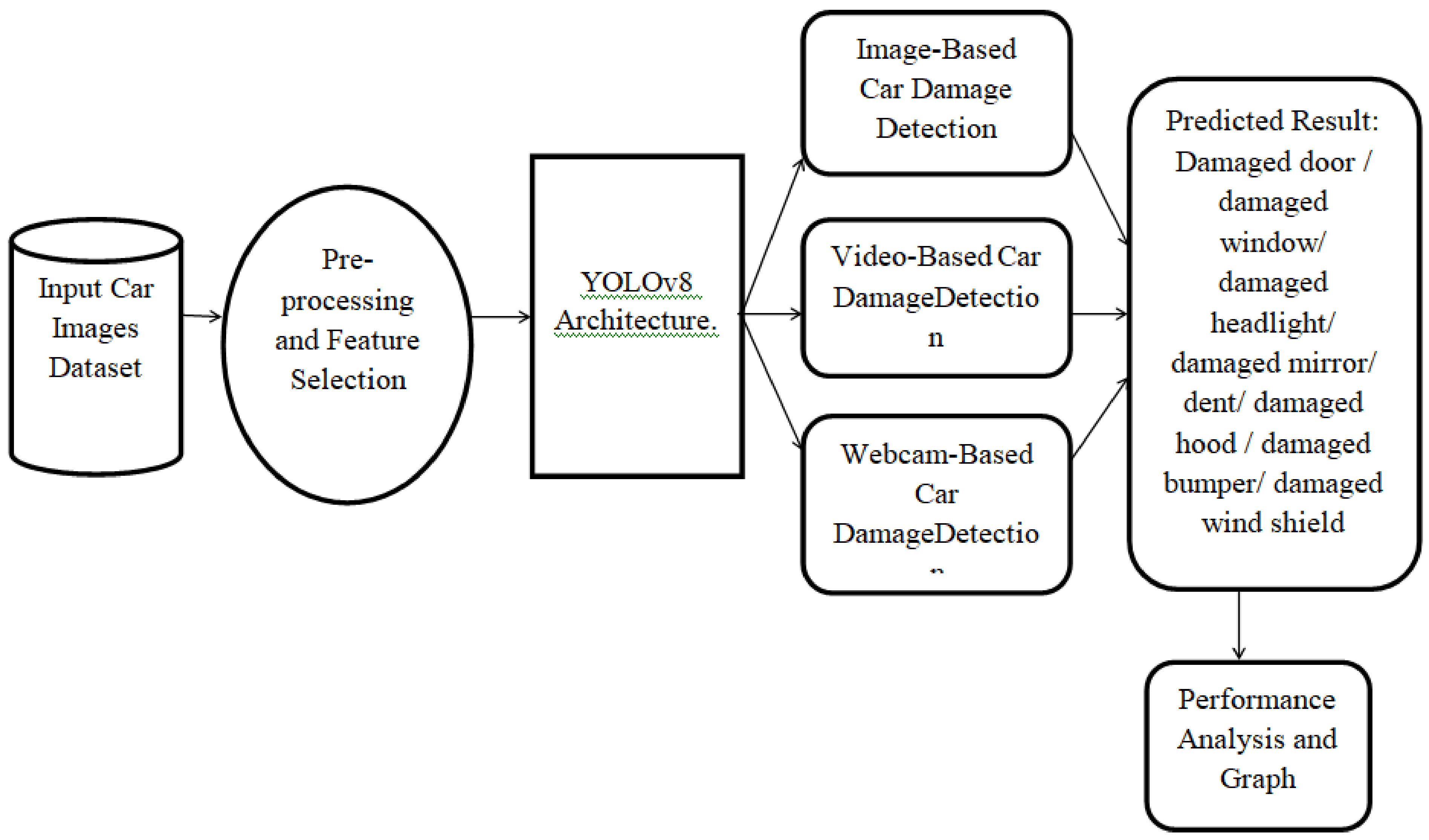
VASAVI COLLEGE OF ENGINEERING (AUTONOMOUS)

INFORMATION TECHNOLOGY

Objective :To develop an automated system using YOLOv8 that detects and localizes vehicle damage from images and videos. The system aims to enhance the speed and accuracy of damage inspection for use in insurance processing, rental services, and fleet management.

Motivation:Manual vehicle damage inspection is time-consuming, subjective, and prone to errors. Automating this process using deep learning enables faster, consistent, and scalable assessments, improving efficiency across industries like insurance, rentals, and fleet management.

PROPOSED SYSTEM ARCHITECTURE:



KEY FEATURES:

Automated detection and localization of vehicle damage from images and videos using YOLOv8.

Supports multiple deployment modes including image upload, video analysis, and web-based interfaces for real-time use.

TECHNOOGIES USED:

- **Deep Learning: YOLOv8 (Ultralytics) for object detection**
- **Programming: Python**
- **Libraries & Tools: OpenCV, PyTorch, Flask**
- **Deployment: Web interface, Video processing, Image analysis**

Team Members:

- **N. Hemantha Giri**
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